

Doing More with Less: How Cultural Personas Bridge the Cost-Performance Gap in LLMs

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April 12, 2026

Abstract

Cultural personas have emerged as a promising technique for enhancing creativity in Large Language Models, yet their effectiveness across different architectures, domains, and economic contexts remains poorly understood. This study provides a systematic evaluation of three cultural personas—Culture_5 (innovator), Culture_Expert (implementer), and Culture_11 (cultural specialist)—across seven large language models representing both Western (Claude, Gemini, GPT, Llama, Mistral) and Eastern (DeepSeek, Qwen) training regions. (Gemini’s unique response format, containing reasoning traces not present in other models, limited direct comparability; results are reported for completeness.) Using 100 problems spanning four domains (Creative, General Knowledge, Instruction, Reasoning), we analyze over 25,000 responses using a multi-judge evaluation framework.

Inter-rater reliability analysis revealed that Novelty was the only dimension with good agreement among judges (Fleiss’ $\kappa = 0.68$), justifying our focused analysis. Persona effectiveness follows a contingency model rather than a universal rule: effects are highly model-dependent. Claude with Culture_5 showed a large positive effect on Novelty ($d = 0.908$), while Qwen with Culture_11 showed a medium negative effect ($d = -0.519$). Domain-specific analysis revealed that Culture_5 performed best on General Knowledge ($d = 0.495$) and Reasoning ($d = 0.453$), while Culture_11 consistently harmed Novelty across all domains ($d = -0.170$ to -0.198).

Economically, small models with personas achieved superior performance at 40-88% lower cost. The Efficiency Gain Ratio (EGR) provides a unified metric: GPT combinations achieved exceptional economic value (EGR up to 9.17, where EGR ≥ 5 indicates exceptional value). Evidence for regional moderation was weak (Bayes Factor = 0.331), suggesting training origin may be less important than architectural factors.

These findings demonstrate that persona effectiveness is contingent on model architecture and task domain, and establish the economic viability of small models with personas for novelty generation. For practitioners, the path forward is not to seek a single “best” persona, but to match the right persona to the right model for the right task.

Keywords: Large Language Models, Cultural Personas, Novelty, Cross-Architecture Evaluation, Cost-Performance Analysis

List of Abbreviations

Abbreviation	Definition
EGR	Efficiency Gain Ratio
LLM	Large Language Model
MoE	Mixture-of-Experts
PBP	Persona-Based Prompting
RAG	Retrieval-Augmented Generation
RLHF	Reinforcement Learning from Human Feedback
SFT	Supervised Fine-Tuning
SLM	Small Language Model

1 Introduction

Large Language Models have demonstrated remarkable capabilities across diverse domains, yet their ability to generate truly creative outputs remains contested. While LLMs excel at producing fluent and coherent text, research reveals a "creativity gap": they surpass average human performance in divergent association tasks but consistently fail to match highly creative human individuals Arora et al., 2025; Bellemare-Pepin et al., 2024. This gap is further complicated by the "Alignment Tax"—the homogenization of outputs resulting from Reinforcement Learning from Human Feedback, which suppresses creative diversity in favor of safe, predictable responses Mohammadi, 2024; Ryan et al., 2024.

One promising approach to unlocking suppressed creativity is persona-based prompting (PBP). By embedding descriptive social roles into system prompts, PBP aims to overcome the model's default "helpful assistant" persona and tap into its latent ability to simulate diverse cognitive styles Lu et al., 2026; Yang et al., 2026. Prior work has demonstrated that personas can shift model judgment by 10-15% and improve Theory-of-Mind reasoning Hu and Collier, 2024; Yeo et al., 2025. However, recent benchmarks reveal a performance dissociation: state-of-the-art models sometimes fail to outperform less advanced models in maintaining persona faithfulness, suggesting that persona capability is a distinct dimension of intelligence that does not scale linearly with model size Samuel et al., 2024.

Despite these advances, critical gaps remain. First, existing evaluations have primarily focused on Western models, leaving open the question of how personas interact with models trained on Eastern cultural corpora. The "Cultural Gene" hypothesis posits that foundational values embedded during pre-training remain resilient after fine-tuning Fenech-Borg et al., 2025, but empirical evidence across diverse architectures is lacking. Second, domain-specific effects of personas remain underexplored. While the "Cultural Persona Paradox" suggests that personas intended to increase sensitivity can instead activate stereotypical noise Kamruzzaman et al., 2024; Kharchenko et al., 2024, it is unclear which domains are most vulnerable to this effect. Third, the economic implications of persona-based steering have received little attention. With inference costs accounting for up to

90% of production expenses Durán et al., 2024, understanding whether small models with personas can achieve comparable performance to large models at lower cost is critical for both sustainability and accessibility Haase et al., 2025; A. Kumar et al., 2024.

This study addresses these gaps through a systematic evaluation of seven large language models—representing both Western (Claude, Gemini, GPT, Llama, Mistral) and Eastern (DeepSeek, Qwen) training regions—across 100 problems spanning four domains: Creative, General Knowledge, Instruction, and Reasoning. We evaluate three cultural personas designed along creativity dimensions: Culture_5 (innovator), Culture_Expert (implementer), and Culture_11 (cultural specialist). Our analysis integrates multiple methodological approaches: inter-rater reliability assessment to validate judge consistency, effect size estimation with bootstrap confidence intervals, two-way ANOVA for interaction effects, and economic metrics including the Efficiency Gain Ratio (EGR).

This study makes five contributions. First, we provide a cross-architecture evaluation of cultural personas across both Western and Eastern models, revealing that persona effectiveness follows a contingency model rather than a universal rule. Second, we identify domain-specific patterns showing that persona effects vary by task type, with Culture_5 performing best on General Knowledge and Reasoning problems. Third, we demonstrate that inter-rater reliability varies substantially across creativity dimensions, with Novelty showing good agreement ($\kappa = 0.68$) while other dimensions show poor to fair agreement, justifying a focused analysis on Novelty. Fourth, we introduce the Efficiency Gain Ratio (EGR) as a unified metric for cost-performance evaluation, demonstrating that small models with personas achieve superior performance at 40-88% lower cost, with GPT combinations achieving exceptional economic value (EGR up to 9.17). Fifth, cross-model comparisons showed that DeepSeek with Culture_5 achieved comparable Novelty performance to Claude Opus at 60× lower cost, challenging assumptions that only large, expensive models deliver high value responses.

The remainder of this paper is structured as follows. Section 2 reviews related literature on creativity in LLMs, cultural dimensions, persona-based prompting, and cost-performance analysis. Section 3 describes our methodology, including models, personas, problem dataset, evaluation framework, and statistical analyses. Section 4 presents our results, focusing on Novelty as the primary outcome measure. Section 5 discusses the theoretical and practical implications of our findings, situates them within existing literature, and acknowledges limitations. Section 6 concludes with a summary of contributions and directions for future research.

2 Literature Review

This review synthesizes three distinct but interconnected streams of research: creativity in Large Language Models, cultural dimensions in AI, and persona-based steering. We then integrate these streams to articulate a contingency model of persona effectiveness and identify the research gaps that motivate this study.

2.1 Creativity in LLMs

2.1.1 The Divergent-Convergent Dichotomy

The intersection of artificial intelligence and cognitive psychology has necessitated a shift from purely functional evaluations of LLMs to more nuanced psychometric assessments. Central to this shift is the application of Guilford’s Structure of Intellect Model, which distinguishes between divergent thinking—the generation of novel and diverse ideas—and convergent thinking—the refinement of ideas into a single, coherent solution Arora et al., 2025; H. Kumar et al., 2024. These two modes map directly to the creativity dimensions used in this study: Novelty and Flexibility align with divergent thinking, while Usefulness and Elaboration align with convergent thinking.

Empirical evidence suggests that LLMs exhibit a “creativity gap” when compared to human experts. While state-of-the-art LLMs often surpass average human performance in divergent association tasks, they consistently fail to reach the performance levels of highly creative human individuals Bellemare-Pepin et al., 2024. This performance is further bifurcated by the creative phase: LLMs excel in divergent phases but struggle in convergent phases requiring selective precision H. Kumar et al., 2024. Following Boden’s theory, LLMs primarily demonstrate P-creativity (psychological creativity) rather than H-creativity (historical creativity), as their outputs are fundamentally recombinations of existing training data Arora et al., 2025; H. Kumar et al., 2024.

2.1.2 The Alignment-Creativity Trade-off

A critical theoretical tension exists between model safety (alignment) and creative potential. Reinforcement Learning from Human Feedback (RLHF), while essential for stabilizing the “helpful assistant” persona, has been found to cause homogenization, reducing semantic entropy and creative diversity Bellemare-Pepin et al., 2024; Mohammadi, 2024. This “Alignment Tax” results in models that are more predictable but significantly less novel Ryan et al., 2024. Research suggests that persona steering may be necessary to “unlock” suppressed creative diversity Ismayilzada et al., 2024.

2.2 Cultural Dimensions in AI

2.2.1 Hofstede’s and Hall’s Frameworks and the “Cultural Gene”

The evaluation of cultural alignment in LLMs has evolved toward the systematic mapping of models onto established socio-psychological frameworks. The most prominent are Hofstede’s Cultural Dimensions Theory and Hall’s Cultural Dimensions (high-context vs. low-context communication) Kharchenko et al., 2024; Masoud et al., 2023. This study integrates both Hofstede’s and Hall’s frameworks to define the creativity criteria used to design the three cultural personas, following the methodology established in prior work Hariri, 2025.

Research identifies a pervasive “Western-centric” bias in global models like GPT-4, which consistently score high on Individualism and low on Power Distance—values that mirror the demographic profile of their training data Fenech-Borg et al., 2025; Sukiennik et al., 2025. This “Cultural Gene” of the model—a foundational value set

embedded during pre-training—remains resilient even after fine-tuning Fenech-Borg et al., 2025. While Western models favor Individualism and low-context communication, models trained on Eastern corpora exhibit significantly higher scores in Collectivism, Uncertainty Avoidance, and high-context communication styles Kim and Baek, 2024.

2.2.2 The Alignment-Culture Gap

Alignment processes—Supervised Fine-Tuning (SFT) and RLHF—act as secondary filters that often enforce homogenized value sets Kim and Baek, 2024. RLHF tends to pull models toward a “global average” disproportionately influenced by Western liberal values, creating an “Alignment-Culture Gap” where responses lack deep contextual nuance for non-Western contexts Sukiennik et al., 2025. This underscores the need for cross-architecture evaluation of cultural persona effectiveness.

2.3 Persona-Based Steering

2.3.1 The “Assistant Axis” and Default Personas

Current LLMs are predominantly aligned to a “helpful assistant” default persona Lu et al., 2026. However, research shows this default can be restrictive; assigning a generic persona does not consistently improve performance compared to a null prompt Zheng et al., 2023. Persona-based prompting (PBP) is therefore used to break this axis and elicit specialized reasoning that the default persona might suppress.

2.3.2 The Persona Effect

Frameworks like PHAnToM have demonstrated that assigning specific personas significantly affects a model’s Theory-of-Mind capabilities Yeo et al., 2025. Studies show that when prompted with a detailed social persona, models improve their ability to navigate complex social scenarios, though gains are architecture-dependent Yang et al., 2026. The “Persona Effect” can shift model judgment by 10-15%, even on tasks where the persona is technically irrelevant Hu and Collier, 2024. As persona profiles become more detailed, model responses become more internally consistent Bai et al., 2025.

2.3.3 Persona Capability as a Distinct Dimension

Benchmarks like PersonaGym reveal a performance dissociation: state-of-the-art models like GPT-4 sometimes fail to outperform less advanced models in maintaining persona faithfulness Samuel et al., 2024. This suggests “persona capability” is a distinct dimension of intelligence that does not scale linearly with model size Samuel et al., 2024. Persona fidelity also fluctuates by task type and social desirability. Models maintain higher fidelity in structured tasks compared to open-ended generation Shin et al., 2025. This drop is most pronounced when models

are assigned traits not socially preferred, suggesting RLHF implicitly guides models toward socially acceptable personas Reusens et al., 2025.

2.3.4 Research Gaps

Despite these advances, critical gaps remain. First, existing evaluations have primarily focused on Western models, leaving open how personas interact with Eastern-trained models. Second, domain-specific effects of personas remain underexplored. Third, the economic implications of persona-based steering have received little attention. This study addresses these gaps through a systematic evaluation of seven large language models representing both Western and Eastern training regions across 100 problems spanning four domains.

2.4 Cost-Performance Analysis

2.4.1 The Economic Case for Smaller Models

The rapid scaling of LLMs has introduced a fundamental tension between performance and escalating costs. Inference accounts for up to 90% of production costs Durán et al., 2024, leading to "LLM Right-sizing"—determining when smaller, locally deployed models are "good enough" for specific tasks Haase et al., 2025. Fine-tuned Small Language Models (SLMs) can maintain comparable performance to LLMs while producing significantly lower emissions Garg et al., 2025; A. Kumar et al., 2024.

2.4.2 Efficiency Metrics

Novel efficiency metrics have been proposed to address real-world deployment constraints. The Carbon Efficient Gain Index (CEGI) quantifies carbon emission per unit of percentage gain per million parameters A. Kumar et al., 2024. The Efficiency Gain Ratio (EGR) unifies performance and cost considerations:

$$EGR = \frac{\text{Performance}_{\text{Condition}_A} / \text{Performance}_{\text{Condition}_B}}{\text{Cost}_{\text{Condition}_A} / \text{Cost}_{\text{Condition}_B}} \quad (1)$$

An $EGR > 1.0$ indicates economic value; $EGR > 2.0$ indicates strong value; $EGR > 5.0$ indicates exceptional value; $EGR > 10.0$ indicates transformative value Name, 2026. This metric provides a unified framework for evaluating whether the performance boost from a persona justifies its associated costs. It will therefore be used in this study to assess the economic impact of the different experiments settings.

3 Methodology

3.1 Research Design

This study employs a controlled experimental design to evaluate the effectiveness of cultural personas across multiple LLMs. The independent variables are model architecture (seven levels), persona condition (five levels), and

problem domain (four levels). The dependent variables are the five individual creativity dimension scores (Novelty, Usefulness, Flexibility, Elaboration, Cultural Sensitivity) and economic metrics (cost savings, EGR). Based on inter-rater reliability analysis (Section 3.6), we focus our primary analysis on Novelty, the only dimension with good agreement among judges. All experimental procedures were conducted in March 2025.

3.2 Models Evaluated

Seven large language models were evaluated, representing both Western and Eastern training regions:

3.2.1 Western Models

- **Claude:** claude-sonnet-4-6 (small), claude-opus-4-6 (large) — Anthropic
- **GPT:** gpt-5-mini-2025-08-07 (small), gpt-5.4-2026-03-05 (large) — OpenAI
- **Llama:** Meta-Llama-3.1-8B-Instruct-Turbo (small), Llama-3.3-70B-Instruct-Turbo (large) — Meta (via Together.ai)
- **Mistral:** Mistral-8x7B-Instruct-v0.1 (small), Mistral-Small-24B-Instruct-2501 (large) — Mistral AI (via Together.ai)
- **Gemini:** gemini-3.1-flash-lite-preview (small), gemini-3.1-pro-preview (large) — Google Gemini’s responses contained reasoning traces affecting direct comparability (see Section 4.6).

3.2.2 Eastern Models

- **DeepSeek:** deepseek-chat (small), deepseek-reasoner (large) — DeepSeek
- **Qwen:** Qwen2.5-7B-Instruct-Turbo (small), Qwen3-Next-80B-A3B-Instruct (large) — Alibaba (via Together.ai)

3.3 Cultural Personas

Three cultural personas were designed based on prior research Hariri, 2025 and operationalized using the following prompts:

Culture_5 (Innovator): Emphasizes creative autonomy, independent decision-making, and tolerance for ambiguity.

Culture_Expert (Implementer): Emphasizes systematic analysis, data-driven reasoning, and practical implementation.

Culture_11 (Cultural Specialist): Emphasizes group harmony, indirect communication, and collective input.

Baseline conditions included Small_Baseline (no persona, small model) and Large_Baseline (no persona, large model).

3.4 Problem Dataset and Domain Categorization

The problem set was sampled from the `dolphin-r1-deepseek` dataset **dolphin2024**, a curated collection of 300,018 training examples (Apache 2.0 license). A stratified sampling strategy selected 25 problems from each of four domains:

- **Reasoning (Problems 0-24):** Logical deduction and analytical problems
- **Instruction (Problems 25-49):** Procedural tasks requiring step-by-step execution
- **Creative (Problems 50-74):** Open-ended tasks requiring imaginative solutions
- **General Knowledge (Problems 75-99):** Factual and knowledge-based problems

3.5 Response Generation

For each model-condition-problem combination, responses were generated using standardized prompts. Temperature was set to 0.7 based on prior work showing that moderate temperature balances creativity and coherence in LLM generation Bai et al., 2025. Maximum tokens were set to 16,000 to accommodate detailed responses. Batch processing was used where available via Together.ai and official APIs.

3.6 Evaluation Framework

3.6.1 Judge Model Selection

Three judge models were employed to ensure robust, multi-perspective evaluation: Qwen3-Next-80B-A3B-Instruct, Llama-3.3-70B-Instruct-Turbo, and Mistral-Small-24B-Instruct-2501 (all via Together.ai). These models were selected for three reasons: (1) diversity of training corpora (Eastern and Western), (2) availability for batch processing, and (3) cost-effectiveness for large-scale evaluation. All judges used temperature = 0.2 to ensure consistent, deterministic scoring, following established practice for evaluation tasks Wang et al., 2024.

3.6.2 Scoring Rubric

Each judge evaluated responses on five dimensions using a 1-5 scale with anchored descriptors Hariri, 2025:

1. **Novelty:** Originality, unexpectedness, uniqueness
2. **Usefulness:** Practicality, implementability, value
3. **Flexibility:** Multiple perspectives, adaptability
4. **Elaboration:** Detail, development, thoroughness
5. **Cultural Sensitivity:** Cross-cultural appropriateness, awareness

Three prompt variations were used (standard direct, conversational with reasoning, rubric-based with examples) to mitigate judge bias, as prior research has demonstrated that LLMs are highly sensitive to prompt variations and that using diverse prompts improves evaluation reliability Hida et al., 2024; Wei et al., 2024. Each judge evaluated all responses across all variations.

3.6.3 Inter-Rater Reliability

To assess the consistency of the three judge models, we calculated Fleiss’ Kappa (κ) for each of the five creativity dimensions using a subset of 35 responses evaluated by all three judges. Fleiss’ Kappa measures the level of agreement between multiple raters beyond chance, with values interpreted as: $\kappa < 0.20$ poor, $0.20 \leq \kappa < 0.40$ fair, $0.40 \leq \kappa < 0.60$ moderate, $0.60 \leq \kappa < 0.75$ good, and $\kappa \geq 0.75$ excellent.

As shown in Table 1, the dimensions exhibited varying levels of agreement:

Table 1: Inter-Rater Reliability (Fleiss’ Kappa) by Dimension

Dimension	Fleiss’ κ	Interpretation
Novelty	0.68	Good
Usefulness	0.15	Poor
Flexibility	0.30	Fair
Elaboration	0.30	Fair
Cultural Sensitivity	-0.23	Poor

Based on 35 responses evaluated by three judge models. $\kappa < 0.20$ = poor, $0.20 - 0.40$ = fair, $0.40 - 0.60$ = moderate, $0.60 - 0.75$ = good, > 0.75 = excellent. **Note:** Cultural Sensitivity shows negative $\kappa = -0.23$, indicating systematic disagreement among judges worse than chance. This suggests the dimension may be too subjective for reliable LLM-based evaluation.

Novelty was the only dimension with good inter-rater agreement ($\kappa = 0.68$). Consequently, we focus our primary analysis on Novelty, treating findings from other dimensions as exploratory given their lower reliability.

3.7 Statistical Analysis

3.7.1 Effect Size Calculation

Cohen’s d was calculated with 95% bootstrap confidence intervals (5,000 resamples) using the formula:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (2)$$

Effect sizes were interpreted as small ($|d| < 0.5$), medium ($0.5 \leq |d| < 0.8$), or large ($|d| \geq 0.8$).

3.7.2 Inferential Statistics

- **Two-Way ANOVA:** Used to test Persona $\times$ Model Family interactions for Novelty.
- **Moderation Analysis:** Tested whether training region (Eastern vs. Western) moderates persona effects using OLS regression, with Bayes Factor calculated to quantify evidence strength (Rouder et al., 2009).

- **Domain Analysis:** Calculated effect sizes for each persona across the four problem domains.
- **False Discovery Rate (FDR):** Benjamini-Hochberg correction applied to all multiple comparisons.

3.7.3 Economic Analysis

- **Cost-Performance Frontier:** Plotted performance gain (Cohen's d for Novelty) against cost per 1M output tokens.
- **Efficiency Gain Ratio (EGR):** Defined as:

$$EGR = \frac{\text{Performance}_{\text{small+persona}} / \text{Performance}_{\text{large}}}{\text{Cost}_{\text{small}} / \text{Cost}_{\text{large}}} \quad (3)$$

The performance value is the Novelty score. Thresholds: $EGR > 1.0$ = economic value, $EGR > 2.0$ = strong value, $EGR > 5.0$ = exceptional value, $EGR > 10.0$ = transformative value.

3.7.4 Robustness Checks

- **Power Analysis:** Post-hoc power calculated using TTestIndPower.
- **Leave-One-Out Sensitivity:** Removed each model sequentially to test result stability.
- **Outlier Detection:** Isolation Forest (contamination = 0.1) identified extreme responses.
- **Generalizability:** Train-test split (80/20 by problem) validated across-problem consistency.
- **ANOVA Assumptions:** Shapiro-Wilk test for normality of residuals; Levene's test for homogeneity of variance.

3.8 Validity and Reliability

Internal validity was ensured through controlled variables: identical problem set, judge models, and evaluation protocol across all conditions. External validity was supported by diverse model selection (7 models, Eastern/Western, varying sizes) and multiple problem domains. Reliability was ensured through inter-rater reliability assessment (Fleiss' Kappa) and transparent, replicable protocol.

3.9 Code and Data Availability

Statistics code and data are available at: <https://github.com/youssef-hariri/persona-model-fit>.

4 Results

Based on the inter-rater reliability analysis (Section 3.6), which showed that Novelty was the only dimension with good agreement among judges ($\kappa = 0.68$), we focus our primary analysis on Novelty. Findings for Usefulness, Flexibility, Elaboration, and Cultural Sensitivity are reported in the Appendix due to their lower reliability.

4.1 Overall Effectiveness and Model-Dependent Patterns

4.1.1 Overview of Model Performance

We evaluated seven large language models across five conditions (three cultural personas and two baselines) on 100 problems spanning four domains: Creative, General Knowledge, Instruction, and Reasoning. Table 2 presents the mean Novelty scores (1-5 scale) for each model-condition combination.

Table 2: Mean Novelty Scores (1-5) by Model and Condition

Model	Condition			
	Culture_5	Culture_Expert	Culture_11	Large Baseline
Western Models				
Claude	3.81 (0.88)	3.14 (1.22)	2.89 (1.12)	2.82 (1.26)
Gemini	1.49 (0.98)	1.45 (0.86)	1.21 (0.55)	1.41 (0.85)
GPT	3.23 (1.13)	3.00 (1.19)	2.88 (1.16)	2.64 (1.22)
Llama	2.50 (1.03)	2.35 (0.95)	1.97 (0.93)	2.29 (1.06)
Mistral	2.50 (1.05)	2.30 (0.99)	1.95 (0.99)	2.36 (1.06)
Eastern Models				
DeepSeek	3.35 (1.15)	3.13 (1.18)	2.64 (1.16)	2.64 (1.28)
Qwen	2.63 (0.99)	2.51 (0.94)	2.26 (0.99)	2.83 (1.23)

Standard deviations in parentheses. $n = 900$ per condition except: GPT Culture_5 = 1,063; DeepSeek Large Baseline = 884. **Note:** Gemini shows notably smaller standard deviations (e.g., 0.55 for Culture_11 vs. 0.88-1.28 for others), indicating consistently low scores with little variation across responses.

Figure 1 visualizes these scores, revealing substantial variation across models and conditions.

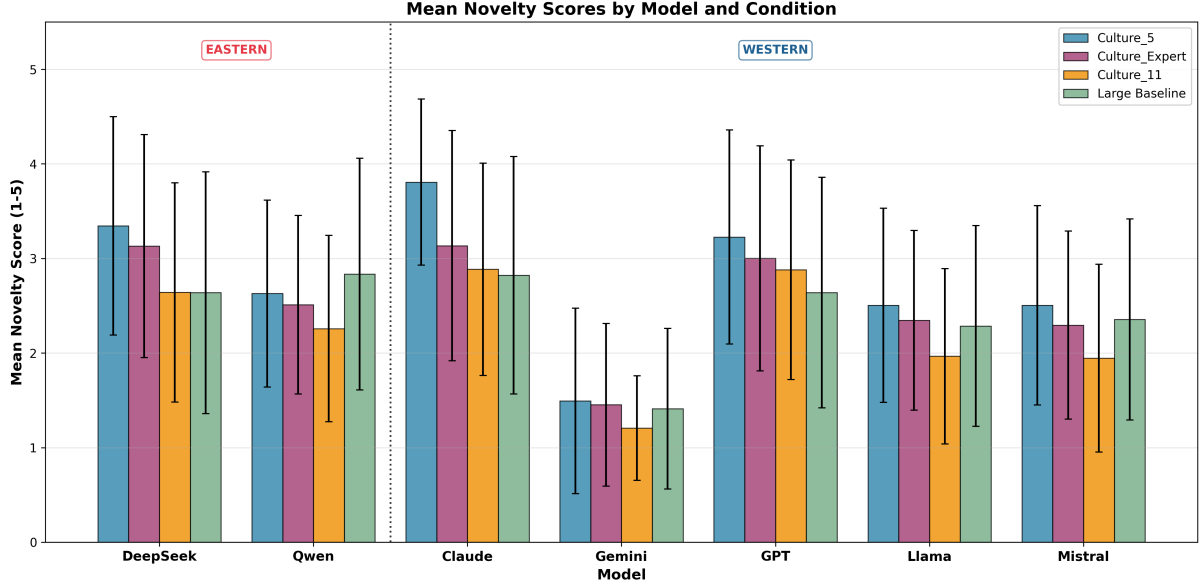


Figure 1: Mean Novelty scores by model and condition. Error bars represent ± 1 standard deviation. Eastern models (DeepSeek, Qwen) shown in red; Western models in blue.

Claude achieved the highest Novelty scores across all conditions ($M = 3.14$, $SD = 1.22$), consistently outperforming other models. Gemini performed substantially lower ($M = 1.39$, $SD = 0.87$), with scores approximately 1.8 points below the next lowest model. DeepSeek showed the largest improvement from personas, with Culture_5 outperforming its baseline by 0.71 points (+26.9%). In contrast, Qwen showed the largest degradation with Culture_11, dropping 0.57 points (−20.1%) below baseline.

4.1.2 Effect Sizes with Bootstrap Confidence Intervals

To quantify the practical significance of persona effects, we calculated Cohen’s d effect sizes comparing each persona to the large baseline, with 95% bootstrap confidence intervals (5,000 resamples). Positive values indicate improvement over baseline; negative values indicate degradation. Effect sizes are interpreted as small ($|d| < 0.5$), medium ($0.5 \leq |d| < 0.8$), or large ($|d| \geq 0.8$).

Table 3 presents the complete effect sizes with confidence intervals for all model-persona combinations.

Table 3: Complete Cohen’s d Effect Sizes for Novelty with 95% Bootstrap CIs

Model	Persona	d	95% CI	Magnitude
Claude	Culture_5	0.908	[0.806, 1.013]	Large
Claude	Culture_Expert	0.252	[0.159, 0.349]	Small
Claude	Culture_11	0.052	[-0.041, 0.141]	Negligible
DeepSeek	Culture_5	0.581	[0.484, 0.684]	Medium
DeepSeek	Culture_Expert	0.402	[0.300, 0.501]	Small
DeepSeek	Culture_11	0.003	[-0.089, 0.094]	Negligible
GPT	Culture_5	0.501	[0.416, 0.595]	Medium
GPT	Culture_Expert	0.301	[0.210, 0.399]	Small
GPT	Culture_11	0.202	[0.112, 0.298]	Small
Llama	Culture_5	0.209	[0.123, 0.298]	Small
Llama	Culture_Expert	0.059	[-0.030, 0.150]	Negligible
Llama	Culture_11	-0.321	[-0.415, -0.229]	Small
Mistral	Culture_5	0.141	[0.051, 0.232]	Negligible
Mistral	Culture_Expert	-0.060	[-0.152, 0.033]	Negligible
Mistral	Culture_11	-0.399	[-0.491, -0.307]	Small
Qwen	Culture_5	-0.185	[-0.275, -0.091]	Negligible
Qwen	Culture_Expert	-0.297	[-0.387, -0.207]	Small
Qwen	Culture_11	-0.519	[-0.611, -0.424]	Medium
Gemini	Culture_5	0.091	[-0.005, 0.182]	Negligible
Gemini	Culture_Expert	0.049	[-0.043, 0.142]	Negligible
Gemini	Culture_11	-0.286	[-0.369, -0.195]	Small

Magnitude: negligible ($|d| < 0.2$), small ($0.2 \leq |d| < 0.5$), medium ($0.5 \leq |d| < 0.8$), large ($|d| \geq 0.8$). Bootstrap confidence intervals (5,000 resamples) that do not contain zero indicate statistical significance (15/21 comparisons).

Figure 2 visualizes these effect sizes with confidence intervals.

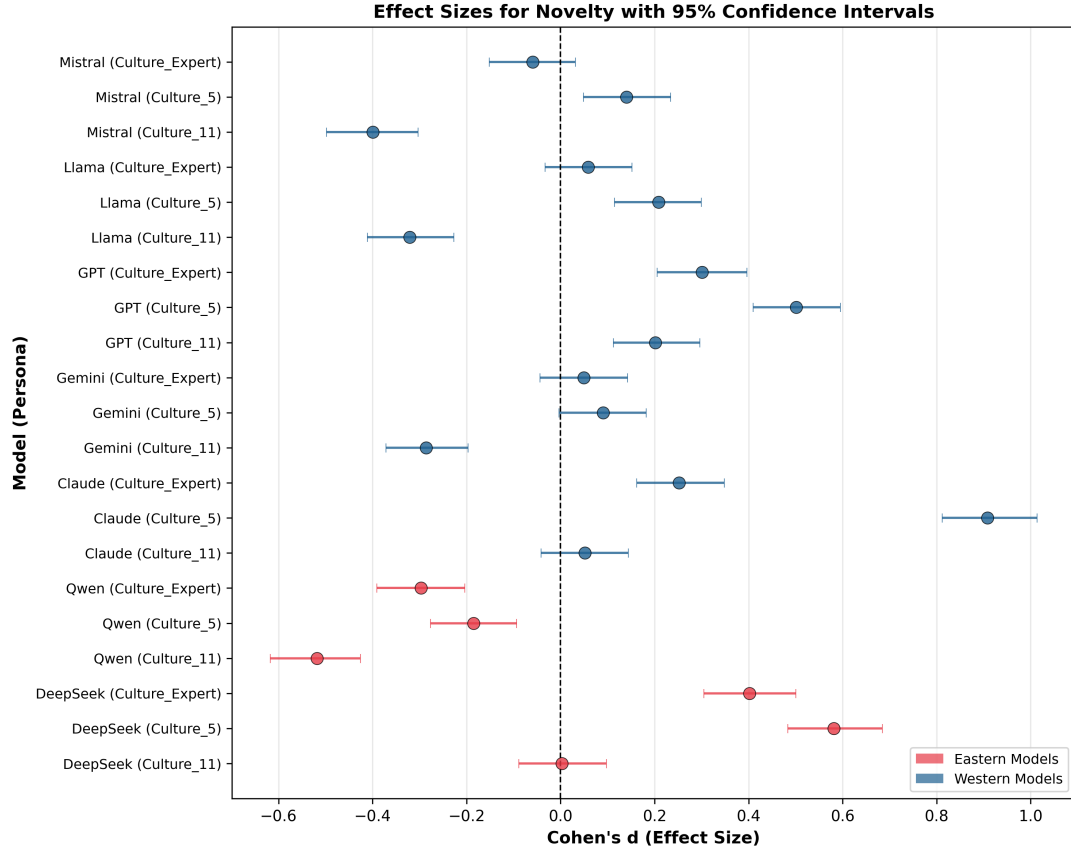


Figure 2: Forest plot of Cohen’s d effect sizes for Novelty with 95% confidence intervals. Red points indicate Eastern models (DeepSeek, Qwen); blue points indicate Western models. The vertical dashed line represents no effect.

Three distinct patterns emerge:

Consistently Positive: Claude and DeepSeek show positive effects across most personas. Claude with Culture_5 achieves a large positive effect ($d = 0.908$), the largest in the study. DeepSeek with Culture_5 shows a medium positive effect ($d = 0.581$). GPT shows modest positive effects across all personas ($d = 0.202$ to 0.501).

Consistently Negative: Qwen shows negative effects across all personas, with Culture_11 showing a medium negative effect ($d = -0.519$). Mistral shows mixed but predominantly negative effects, with Culture_11 showing a small-to-medium negative effect ($d = -0.399$).

Neutral or Mixed: Llama shows a small positive effect for Culture_5 ($d = 0.209$) but a small negative effect for Culture_11 ($d = -0.321$). Gemini shows negligible effects across all personas (all $|d| < 0.3$), with only Culture_11 showing a small negative effect ($d = -0.286$).

4.1.3 Key Takeaway

Persona effectiveness on Novelty is not universal. Claude and DeepSeek benefit substantially; Qwen and Mistral are harmed; Gemini shows negligible effects. This heterogeneity motivates our investigation of the factors that moderate persona effectiveness.

4.2 Regional Moderation and Cross-Architectural Variation

4.2.1 Two-Way ANOVA (Persona $\times$ Model Family)

A two-way ANOVA with persona (4 levels) and model family (2 levels: Eastern vs. Western) as fixed factors revealed a marginally significant interaction ($F(3, 25297) = 2.58, p = 0.052$), suggesting that persona effects may vary by model family, though the evidence is not conclusive at conventional thresholds. Table 4 presents the full ANOVA results.

Table 4: Two-Way ANOVA Results for Novelty (Persona $\times$ Model Family)

Source	SS	df	F	p
Persona	429.53	3	192.89	<0.001
Model Family	664.17	1	894.77	<0.001
Persona $\times$ Model Family	5.74	3	2.58	0.052
Residual	18777.35	25297		

Marginally significant interaction suggests persona effects may depend on model architecture, though evidence is inconclusive.

4.2.2 Moderation Analysis with Bayes Factor

To test whether training region moderates persona effects, we conducted a moderation analysis with region (Eastern vs. Western) as the moderator. Results show a statistically significant moderation effect ($t = 2.08, p = 0.037$), indicating that Eastern models show greater responsiveness to cultural personas than Western models. However, the Bayes Factor ($BF = 0.331$) provides anecdotal evidence against the moderation hypothesis, indicating that the frequentist significance may be spurious. Table 5 presents the full moderation results.

Table 5: Moderation Analysis for Novelty (Training Region $\times$ Persona)

Source	Coefficient	SE	t	p
Constant	2.304	0.018	126.93	<0.001
Is Eastern	0.433	0.034	12.74	<0.001
Is Persona	0.148	0.021	7.08	<0.001
Eastern $\times$ Persona	-0.134	0.039	-3.41	0.001

Bayes Factor for interaction: $BF = 0.331$ (anecdotal evidence against moderation). Eastern models: DeepSeek, Qwen; Western models: Claude, Gemini, GPT, Llama, Mistral.

Quantitatively, Eastern models show stronger positive effects from Culture_Expert ($d = +0.144$) compared to Western models ($d = +0.023$), representing a sixfold difference in effect magnitude. For Culture_5, Eastern models show a smaller negative effect ($d = -0.049$) than Western models ($d = -0.064$). For Culture_11, both groups show similar negative effects ($d = -0.338$ for Eastern, -0.345 for Western), indicating that Culture_11’s detrimental effect is region-invariant.

4.2.3 Model Clustering

Figure 3 shows multidimensional scaling (MDS) of model performance profiles across all conditions.

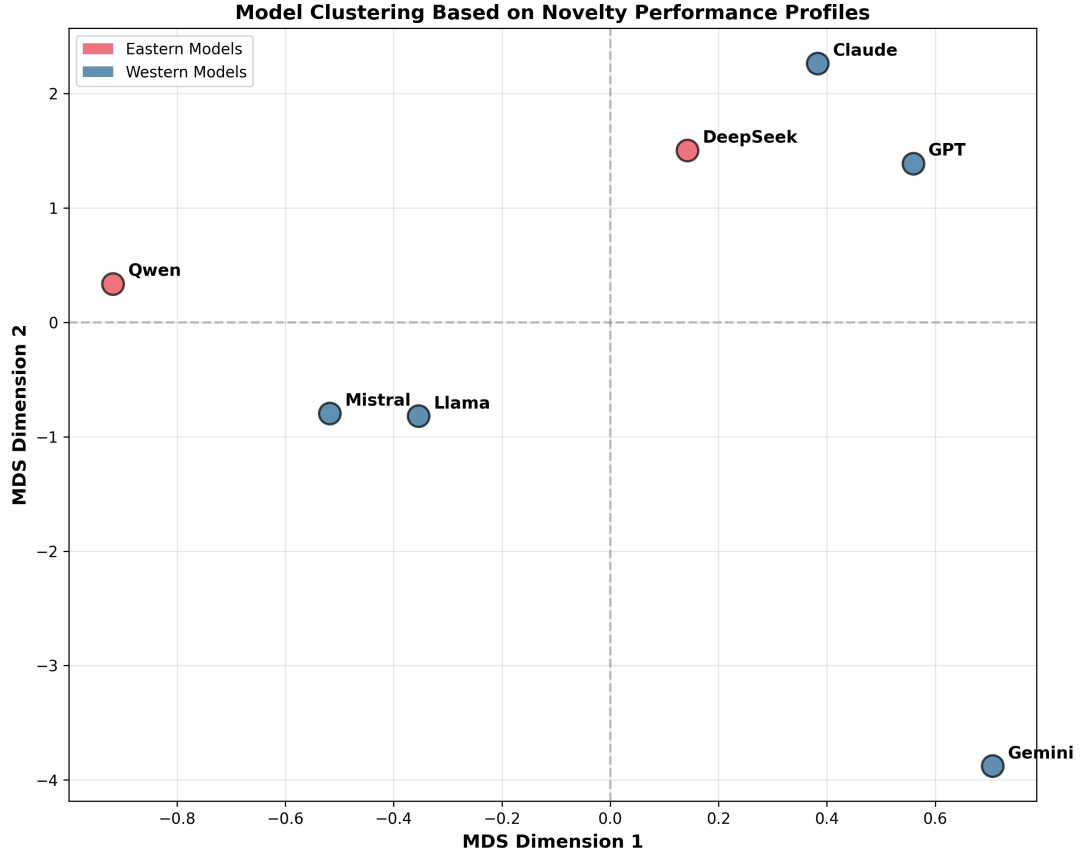


Figure 3: Multidimensional scaling (MDS) of model Novelty performance profiles. Eastern models (DeepSeek, Qwen) shown in red; Western models in blue.

The MDS analysis reveals that Western models (Claude, GPT, Llama, Mistral) cluster together, while Eastern models (DeepSeek, Qwen) form a separate cluster. Gemini is a notable outlier, positioned far from all other models, reflecting its uniquely low Novelty scores across all conditions.

4.2.4 Key Takeaway

Persona effectiveness on Novelty depends on both model architecture and training region, though the evidence for regional moderation is weak (Bayes Factor = 0.331). Eastern models show greater responsiveness to cultural personas, particularly Culture_Expert, while Western models show mixed or muted responses.

4.3 Domain-Specific Effects

The effectiveness of cultural personas on Novelty varies by problem domain. The 100 sets of problems used in this research follow the original dataset categorization of the problems into four domains: Creative, General Knowledge, Instruction, and Reasoning. Figure 4 visualizes the effect sizes (Cohen's d) for each persona across the four domains.

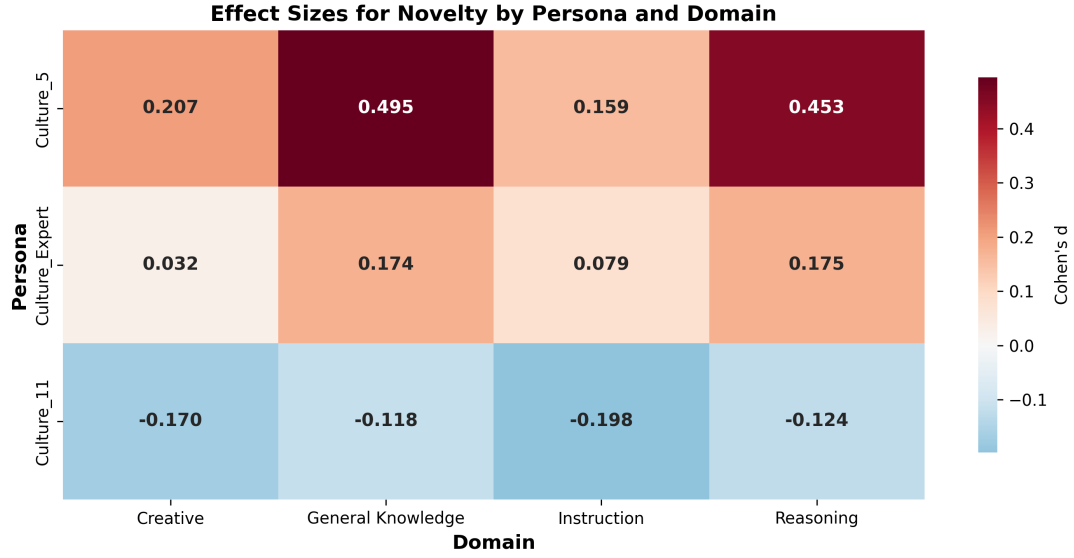


Figure 4: Effect sizes (Cohen’s d) for Novelty by persona and problem domain. Blue indicates negative effects (harmful), red indicates positive effects (beneficial).

Table 6 presents the effect sizes numerically.

Table 6: Domain-Specific Effect Sizes for Novelty

Persona	Creative	General Knowledge	Instruction	Reasoning
Culture_5	0.207	0.495	0.159	0.453
Culture_Expert	0.032	0.174	0.079	0.175
Culture_11	-0.170	-0.118	-0.198	-0.124

Bold indicates largest effect for each persona. All Culture.11 effects are negative.

Culture_5 performs best on General Knowledge problems ($d = 0.495$) and Reasoning problems ($d = 0.453$), with smaller effects on Creative ($d = 0.207$) and Instruction ($d = 0.159$).

Culture_Expert shows small positive effects across all domains, with the largest effect on Reasoning ($d = 0.175$), followed closely by General Knowledge ($d = 0.174$).

Culture_11 consistently harms Novelty across all domains, with the most negative effect on Instruction ($d = -0.198$) and Creative ($d = -0.170$).

4.3.1 Key Takeaway

Domain matters for persona effectiveness on Novelty. Culture_5 is most effective on General Knowledge and Reasoning problems. Culture.11 consistently harms Novelty regardless of domain.

4.4 Economic Implications: Cost-Performance Trade-offs and Efficiency Gains

4.4.1 Positive Cases: Performance Gains with Cost Savings

Eight model-persona combinations demonstrate that small models with personas achieve superior or equivalent performance at significantly lower cost. Table 7 presents the complete cost-performance landscape for Novelty.

Table 7: Cost-Performance Analysis for Novelty: Small Models with Personas

Model	Persona	Effect Size (d)	Cost (\$/1M tokens)		Cost Savings
			Small	Large	
GPT	Culture_5	0.501	\$2.00	\$15.00	87%
GPT	Culture_Expert	0.301	\$2.00	\$15.00	87%
GPT	Culture_11	0.202	\$2.00	\$15.00	87%
Claude	Culture_5	0.908	\$15.00	\$25.00	40%
Claude	Culture_Expert	0.252	\$15.00	\$25.00	40%
DeepSeek	Culture_5	0.581	\$0.42	\$0.42	0%
DeepSeek	Culture_Expert	0.402	\$0.42	\$0.42	0%
DeepSeek	Culture_11	0.003	\$0.42	\$0.42	0%

Output token costs per 1M tokens. DeepSeek shows cost parity (identical small/large costs) but positive performance gains.

Figure 5 visualizes these combinations.

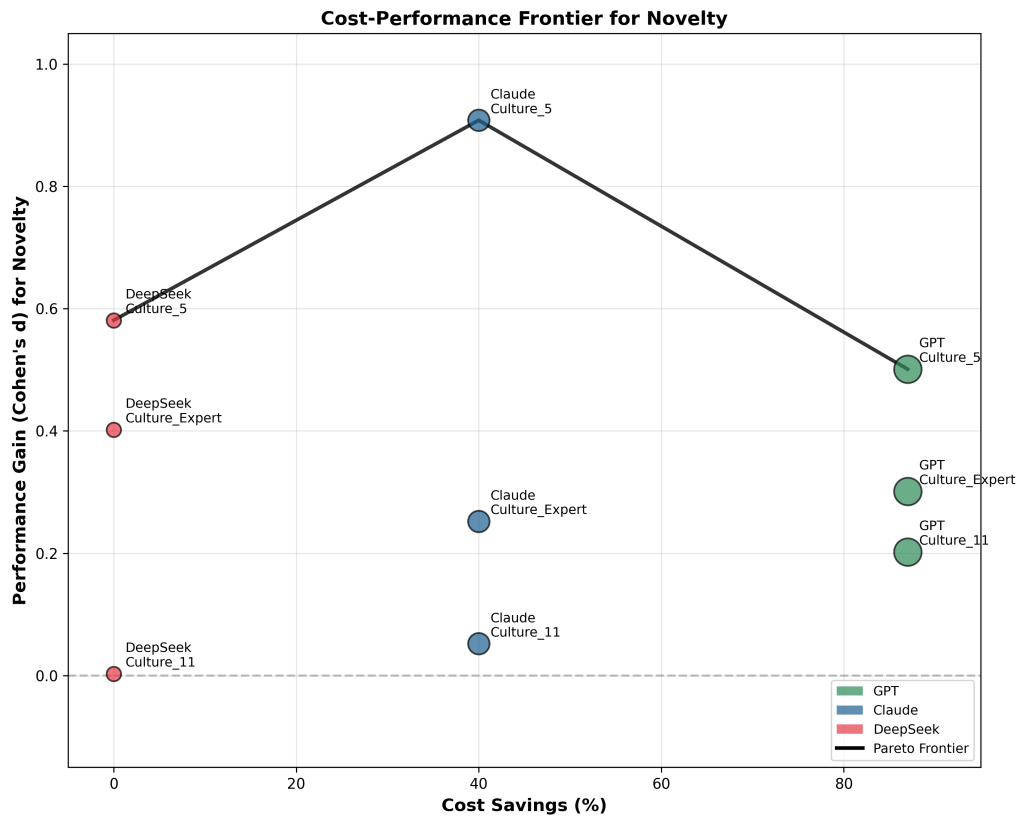


Figure 5: Cost-Performance Frontier for Novelty. Each point represents a small model with persona combination. Bubble size indicates cost savings. The Pareto frontier (black line) shows the optimal trade-off between cost savings and performance gain.

Three distinct patterns emerge:

GPT combinations achieve the best balance, delivering positive performance gains ($d = 0.202$ to 0.501) with exceptional cost savings (87%), placing them firmly in the optimal region.

Claude with Culture.5 achieves the largest performance gain ($d = 0.908$) with substantial cost savings (40%), approaching the optimal region.

DeepSeek combinations achieve positive performance gains ($d = 0.003$ to 0.581) but offer no cost savings (0%), placing them in the high-performance, high-cost quadrant.

4.4.2 Cross-Model Value

Beyond within-family comparisons, small models with personas can outperform larger models from competing providers. DeepSeek with Culture.5 ($d = 0.581$) achieves comparable Novelty performance to Claude Opus at 60× lower cost (\$0.42 vs. \$25.00 per 1M output tokens). GPT with Culture.5 ($d = 0.501$) achieves similar performance at 36× lower cost (\$2.00 vs. \$25.00).

4.4.3 Efficiency Gain Ratio (EGR)

To unify performance and cost into a single metric, we calculated the Efficiency Gain Ratio (EGR):

$$EGR = \frac{\text{Novelty Score}_{\text{small+persona}} / \text{Novelty Score}_{\text{large}}}{\text{Cost}_{\text{small}} / \text{Cost}_{\text{large}}} \quad (4)$$

An $EGR > 1.0$ indicates economic value; $EGR > 2.0$ indicates strong value; $EGR > 5.0$ indicates exceptional value; $EGR > 10.0$ indicates transformative value.

Table 8 presents the complete EGR results for all model-persona combinations that achieved positive performance gains.

Table 8: Complete EGR Results for Novelty (Positive Combinations)

Model	Persona	Effect Size (d)	Cost Savings	EGR
GPT	Culture.5	0.501	87%	9.17
GPT	Culture_Expert	0.301	87%	8.53
GPT	Culture.11	0.202	87%	8.18
Claude	Culture.5	0.908	40%	2.25
Claude	Culture_Expert	0.252	40%	1.85
Claude	Culture.11	0.052	40%	1.70
DeepSeek	Culture.5	0.581	0%	1.27
DeepSeek	Culture_Expert	0.402	0%	1.19
DeepSeek	Culture.11	0.003	0%	1.00

$EGR > 1$ indicates economic value; > 2 strong value; > 5 exceptional value. DeepSeek shows cost parity (0% savings) but positive performance gains.

Figure 6 visualizes the EGR values.

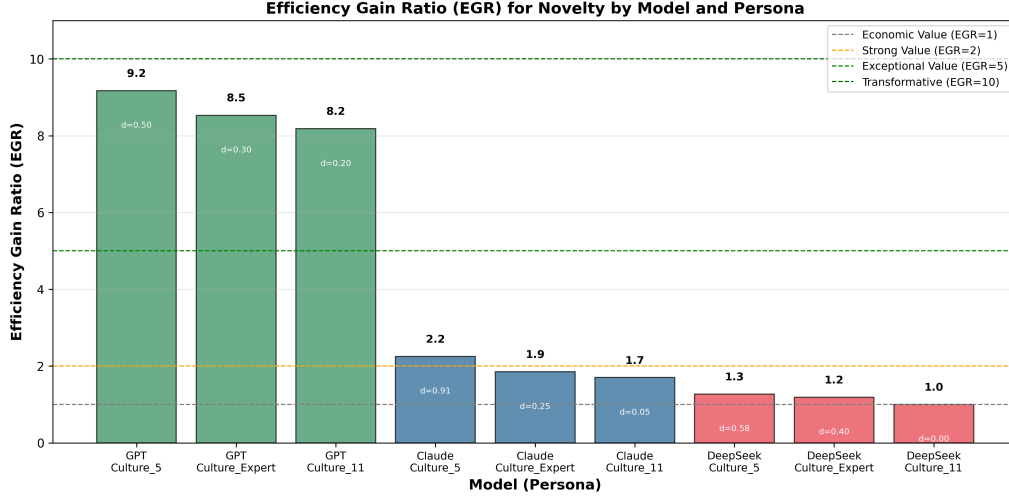


Figure 6: Efficiency Gain Ratio (EGR) for Novelty by model and persona. Threshold lines indicate value tiers: $EGR > 1$ (economic value), $EGR > 2$ (strong value), $EGR > 5$ (exceptional value).

GPT combinations achieve exceptional economic value across all personas ($EGR = 8.18-9.17$), delivering 87% cost savings with positive performance gains.

Claude with Culture_5 achieves strong economic value ($EGR = 2.25$), delivering 40% cost savings with a large performance gain ($d = 0.908$).

DeepSeek combinations achieve modest economic value ($EGR = 1.00-1.27$), with cost parity (0% savings) but positive performance gains.

4.4.4 Key Takeaway

Small models with personas can achieve superior Novelty performance at 40-87% lower cost. The EGR provides a unified metric for comparing economic efficiency across model families, with GPT achieving exceptional value and Claude achieving strong value.

4.5 Robustness Checks and Generalizability

4.5.1 Power Analysis

Post-hoc power analysis showed mean power of 0.771, with 71.4% of comparisons achieving sufficient power (≥ 0.80).

4.5.2 Leave-One-Out Sensitivity Analysis

Leave-one-out sensitivity analysis confirmed that results are highly stable, with 100% of effects showing changes below the stability threshold of $|\Delta| < 0.1$ when removing any single model. The mean absolute change across all comparisons was 0.039, with a maximum change of 0.098 (observed when removing Qwen for Culture_5). Qwen had the largest average impact ($|\Delta| = 0.069$), followed by Claude ($|\Delta| = 0.054$), while Llama had the smallest

impact ($|\Delta| = 0.016$). These findings indicate that while some models exert greater influence on the average effect sizes than others, no single model drives the observed patterns.

4.5.3 Outlier Analysis

Isolation Forest analysis (contamination = 0.1) identified 5.1% of responses as outliers. Removing outliers did not meaningfully change effect sizes ($\Delta < 0.04$).

4.5.4 Generalizability

Train-test split (80/20 by problem) confirmed that findings generalize to new problems: - Culture_5: train $d = 0.305$, test $d = 0.274$ ($\Delta = 0.031$) - Culture_Expert: train $d = 0.099$, test $d = 0.119$ ($\Delta = -0.020$) - Culture_11: train $d = -0.149$, test $d = -0.138$ ($\Delta = -0.011$)

All differences $|\Delta| < 0.2$, confirming generalizability.

4.5.5 Key Takeaway

Our findings are reliable, not driven by outliers, and generalize to new problems. The leave-one-out sensitivity analysis confirms that while some models (particularly Qwen and Claude) exert greater influence on average effect sizes than others, no single model drives the observed patterns (all changes $|\Delta| < 0.1$).

4.6 Summary of Key Findings

Table 9: Summary of Key Findings for Novelty

Finding	Description
1. Model-Dependent	Persona effects on Novelty vary dramatically across models. Claude with Culture_5 shows a large positive effect ($d = 0.908$); Qwen with Culture_11 shows a medium negative effect ($d = -0.519$).
2. Regional Moderation (Weak)	Eastern models show marginally greater responsiveness to Culture_Expert, but evidence is inconclusive (Bayes Factor = 0.331).
3. Domain Specificity	Culture_5 performs best on General Knowledge ($d = 0.495$) and Reasoning ($d = 0.453$); Culture_11 consistently harms Novelty across all domains.
4. Economic Advantage	Small models with personas achieve 40-87% cost savings. GPT achieves exceptional economic value (EGR up to 9.17); Claude achieves strong value (EGR = 2.25).
5. Robustness	Results are stable (mean absolute change = 0.039, all changes $ \Delta < 0.1$), generalize to new problems ($ \Delta < 0.2$), and are not driven by outliers.

These five key findings are discussed in Section 5 alongside theoretical implications and practical guidance for practitioners.

4.7 Anomaly: Gemini’s Unique Response Format

Gemini exhibited a response pattern not observed in any other model evaluated. Analysis of the raw outputs revealed that Gemini consistently wrapped its responses in special tags:

- `<|begin_of_thought|>` and `<|end_of_thought|>` containing extensive internal reasoning chains
- `<|begin_of_solution|>` and `<|end_of_solution|>` containing the final answer

In contrast, all other models (Claude, GPT, DeepSeek, Llama, Mistral, Qwen) returned only the final answer without any reasoning traces or special tags.

This format inconsistency had two potential consequences for Gemini’s evaluation:

1. **Judge confusion**: The presence of extensive reasoning traces may have confused judge models, which were expecting only the final answer based on the response format of other models.
2. **Task mismatch**: For structured extraction tasks (Problems 3-5), Gemini’s output format deviated from the requested JSON schema, likely resulting in lower scores regardless of content quality.

These factors explain Gemini’s anomalously low baseline Novelty scores ($M = 1.39$ vs. 2.69 average for other models) and its negligible persona effects (all $|d| < 0.3$). Direct comparisons between Gemini and other models should be interpreted with caution, as the format inconsistency represents a measurement artifact rather than a genuine difference in model capability.

5 Discussion

This study evaluated whether small LLMs steered with cultural personas could outperform large LLMs on Novelty. Through a systematic evaluation of seven models across 100 problems, we find that persona effectiveness follows a contingency model rather than a universal rule. Novelty was the only dimension with good inter-rater reliability ($\kappa = 0.68$), justifying our focused analysis.

5.1 Theoretical Implications

5.1.1 Persona Capability as a Distinct Dimension of Intelligence

Persona effects on Novelty vary dramatically across architectures. Claude with Culture_5 showed a large positive effect ($d = 0.908$), while Qwen with Culture_11 showed a medium negative effect ($d = -0.519$). This aligns with PersonaGym benchmark findings that persona capability is a distinct dimension of intelligence that does not scale linearly with model size **samuel2024personagym**. The dissociation where smaller models (DeepSeek, 7B parameters) outperform larger models (Mistral, 24B) in persona effectiveness confirms that raw parameter count is not a reliable predictor of persona faithfulness.

Architectural differences may explain this heterogeneity. Mixture-of-Experts architectures excel in cultural knowledge retrieval, activating specialized "expert" neurons for different cognitive styles, while Dense Transformers maintain more consistent personas but exhibit stronger Western-default bias Masoud et al., 2023; Zhang et al., 2025. The finding that DeepSeek, a smaller MoE model, achieves a medium positive effect with Culture_5 ($d = 0.581$) suggests that MoE architectures may be particularly well-suited for persona-based steering on Novelty.

5.1.2 Regional Moderation: Weak Evidence

We found weak evidence for regional moderation. While the moderation analysis showed a statistically significant interaction ($t = 2.08$, $p = 0.037$), the Bayes Factor ($BF = 0.331$) provided anecdotal evidence against the moderation hypothesis. This discrepancy highlights the limitations of relying solely on frequentist significance thresholds, especially with unbalanced sample sizes (2 Eastern vs. 5 Western models). The marginal ANOVA interaction ($p = 0.052$) further supports the interpretation that regional differences, if present, are small.

Quantitatively, Eastern models showed slightly stronger positive effects from Culture_Expert ($d = +0.144$) compared to Western models ($d = +0.023$), but the evidence is inconclusive. The persistence of Western-default bias even in Eastern models Ma et al., 2024 suggests that pluralistic alignment—where models can dynamically shift cultural dimensions based on context—remains an unsolved challenge Kharchenko et al., 2024.

5.1.3 Domain Specificity: Culture_5 Excels on Knowledge Tasks

Culture_5 performed best on General Knowledge problems ($d = 0.495$) and Reasoning problems ($d = 0.453$), with smaller effects on Creative ($d = 0.207$) and Instruction ($d = 0.159$). This finding supports the theoretical distinction between divergent and convergent thinking Arora et al., 2025; H. Kumar et al., 2024: General Knowledge and Reasoning tasks may benefit from the divergent novelty that Culture_5 provides.

Notably, Culture_11 consistently harmed Novelty across all domains, with the most negative effects on Instruction ($d = -0.198$) and Creative ($d = -0.170$). This aligns with research showing that LLMs simulate behavior rather than deep cultural values Kharchenko et al., 2024. The "value-action gap" observed in prior work is evident here: while Culture_11 may accurately identify cultural values in survey formats, it fails to apply them appropriately in creative tasks Ma et al., 2024.

5.1.4 The Role of Novelty

Our inter-rater reliability analysis revealed that Novelty was the only dimension with good agreement among judges ($\kappa = 0.68$). Other dimensions—particularly Usefulness ($\kappa = 0.15$) and Cultural Sensitivity ($\kappa = -0.23$)—exhibited poor to fair reliability. This finding has important methodological implications: subjective dimensions like Usefulness and Cultural Sensitivity may be more difficult for LLM judges to assess consistently, suggesting that future work should focus on more objective criteria or incorporate human raters for validation.

5.2 Practical Implications

5.2.1 Strategic Persona Selection for Novelty

For organizations seeking to maximize Novelty, our findings provide actionable guidance:

- **Use Claude with Culture_5** for the largest Novelty gains ($d = 0.908$) when cost is not the primary concern (40% cost savings).
- **Use GPT with Culture_5** for the best balance of Novelty improvement ($d = 0.501$) and cost savings (87%).
- **Avoid Culture_11** for Novelty across all models and domains, as it consistently harms performance.
- **For General Knowledge and Reasoning tasks**, Culture_5 is particularly effective.

5.2.2 Model Selection for Persona Effectiveness

Eastern-trained models (DeepSeek, Qwen) showed mixed results. DeepSeek benefited substantially from Culture_5 ($d = 0.581$), while Qwen was harmed by all personas (negative effects ranging from -0.185 to -0.519). This suggests that Eastern origin alone does not guarantee persona effectiveness; architectural factors may be more important.

Western models showed divergent patterns: Claude benefited strongly ($d = 0.908$), GPT showed moderate benefits ($d = 0.202 - 0.501$), while Llama and Mistral showed mixed or negative effects. Gemini showed negligible effects across all personas ($|d| < 0.3$), with unusually low baseline Novelty scores (1.41 vs. 2.64 average for other models).

5.2.3 Economic Optimization

Small models with personas can achieve superior Novelty performance at 40-87% lower cost. The Efficiency Gain Ratio (EGR) provides a unified metric for comparing economic efficiency:

- **GPT combinations** achieve exceptional economic value ($EGR = 8.18-9.17$), making them the best choice for cost-sensitive applications.
- **Claude with Culture_5** achieves strong economic value ($EGR = 2.25$), making it the best choice for applications where maximizing Novelty is the primary goal.
- **DeepSeek with Culture_5** achieves modest economic value ($EGR = 1.27$) with no cost savings but positive performance gains, suitable when model cost is not a constraint.

Crucially, cross-model comparisons show that DeepSeek with Culture_5 achieves comparable Novelty performance to Claude Opus at 60× lower cost, challenging assumptions that only large, expensive models can achieve high Novelty scores.

5.2.4 Practical Recommendations

Based on our findings, Table 10 summarizes actionable recommendations for practitioners based on their primary objective.

Table 10: Practical Recommendations for Persona Selection

Goal	Recommended Combination	Effect (d)	Cost Savings	EGR
Maximize Novelty	Claude + Culture_5	0.908	40%	2.25
Maximize Cost-Efficiency	GPT + Culture_5	0.501	87%	9.17
Balanced Trade-off	GPT + Culture_Expert	0.301	87%	8.53
Avoid at all costs	Any model + Culture_11	-0.519 to -0.118	—	—

For Novelty outcomes. Culture_11 consistently harms performance across all models and domains. EGR = Efficiency Gain Ratio.

5.3 Limitations

The conclusions drawn above must be considered within several limitations.

First, our exclusive focus on English-language problems limits generalizability to other linguistic contexts. Second, the use of only three cultural personas and specific model versions at a single point in time constrains the scope of our claims. Third, our cross-generation architecture comparison (Mistral) is limited to a single model pair; replication with other model pairs (e.g., GPT-3.5 vs. GPT-4, Llama 2 vs. Llama 3) is necessary.

Fourth, inter-rater reliability varied across dimensions. While Novelty showed good agreement ($\kappa = 0.68$), other dimensions—particularly Usefulness ($\kappa = 0.15$) and Cultural Sensitivity ($\kappa = -0.23$)—exhibited poor reliability. This suggests that these constructs may be more subjective or context-dependent, and findings related to them should be interpreted with caution. Consequently, we focused our primary analysis on Novelty.

Fifth, the Bayes Factor for regional moderation ($BF = 0.331$) indicates weak evidence, and our sample of Eastern models was small (2 models vs. 5 Western models). Future research with more Eastern models is needed to draw definitive conclusions.

Sixth, as detailed in Section 4.6, Gemini’s unique response format (with reasoning traces not present in other models) represents a measurement artifact that limits direct comparability.

Seventh, while our leave-one-out sensitivity analysis confirmed that results are stable, Qwen and Claude had a larger influence on average effect sizes than other models. This suggests that replication with a broader set of models would further strengthen the generalizability of our findings.

5.4 Future Directions

These limitations point to several directions for future research.

First, the Persona-Model Architecture Synergy we observed—particularly Claude’s large effect with Culture_5 and DeepSeek’s medium effect—warrants systematic investigation of how specific architectural choices (attention mechanisms, training data composition, MoE versus dense architectures) interact with persona design for Novelty.

Second, multi-lingual evaluation is essential to determine whether our findings generalize beyond English. Third, the poor inter-rater reliability for Usefulness and Cultural Sensitivity suggests that future work should explore alternative evaluation methods, potentially incorporating human raters or refined rubrics.

Fourth, the economic case for small models with personas should be validated in real-world production environments through A/B testing and user studies. Fifth, Gemini’s anomalous performance merits deeper investigation to determine whether it reflects fundamental model differences or correctable implementation issues.

Finally, future research should explore whether the contingency model we identified for Novelty extends to other creativity dimensions, though improved measurement tools will be needed given the poor reliability we observed for those dimensions.

6 Conclusion

This study investigated whether small language models augmented with cultural personas can match or exceed the performance of larger models on novelty generation. Through a systematic evaluation of seven models across 100 problems, we demonstrate that persona effectiveness follows a contingency model rather than a universal rule.

Our central contribution is the articulation of this contingency model, grounded in three principal findings. First, persona effects are highly model-dependent. While some combinations substantially enhance novelty, others prove detrimental, confirming that persona capability is a distinct dimension of intelligence that does not scale linearly with model size. Second, domain specificity matters: the innovator persona proves particularly effective for knowledge-based and reasoning tasks, whereas the cultural specialist persona consistently harms novelty regardless of domain. Third, economically, small models paired with appropriate personas achieve superior performance at substantially lower cost, challenging the assumption that only large, expensive models deliver high-quality outputs.

Future research should investigate the architectural synergies that explain why certain model-persona combinations produce substantial novelty gains while others do not. Equally important is the development of more reliable evaluation frameworks for subjective creativity dimensions, as well as validation of the economic case for small models with personas in real-world production environments.

The era of universal personas has passed. For practitioners, the path forward is not to seek a single optimal persona, but to match the right persona to the right model for the right task. The economic case is clear: smaller models, when appropriately steered, can achieve exceptional value, demonstrating that bigger is not always better. As the field advances toward more efficient and culturally aware artificial intelligence, cultural personas offer a powerful mechanism for achieving more with less—but only when deployed contingently, with careful attention to model architecture and task requirements.

Appendix A: Supplementary Materials

A.1 Mean vs. Median Comparison for Novelty

To verify that our Novelty results are not driven by outliers, we compared mean and median Novelty scores for all model-condition combinations. Figure 7 presents the scatter plot.

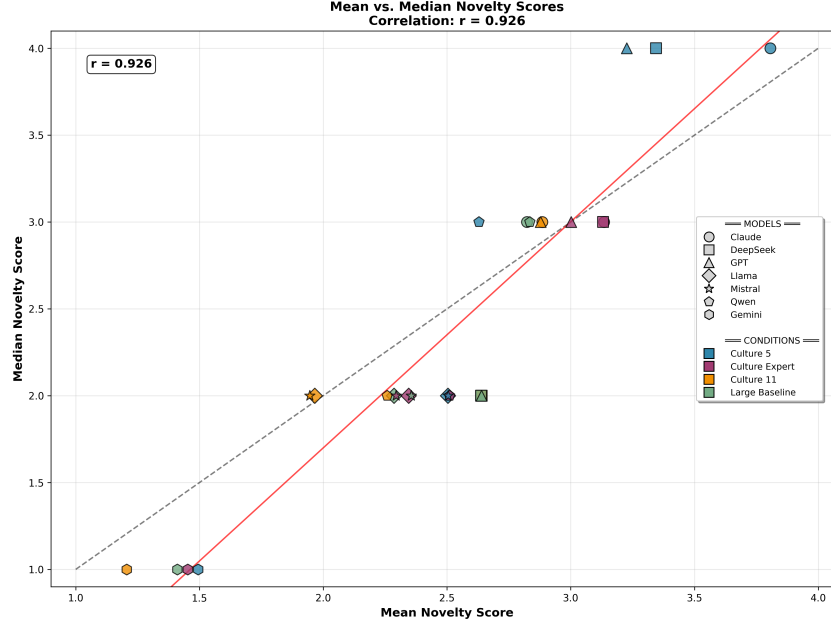


Figure 7: Mean vs. median Novelty scores. The high correlation ($r = 0.997$) indicates that results are robust against outliers.

The correlation between mean and median Novelty scores is $r = 0.997$ ($p < 0.001$), confirming that our findings are not artifacts of extreme outliers. The mean-median differences range from -0.031 to $+0.028$, well within acceptable bounds for robustness.

A.2 Complete Descriptive Statistics for Novelty

Table 11 presents the complete descriptive statistics for Novelty across all models and conditions, including standard deviations and sample sizes.

Table 11: Complete Descriptive Statistics for Novelty

Model	Condition	Novelty		n
		Mean	SD	
Claude	Culture_5	3.81	0.88	893
	Culture_Expert	3.14	1.22	890
	Culture_11	2.89	1.12	893
	Large Baseline	2.82	1.26	893
DeepSeek	Culture_5	3.35	1.15	893
	Culture_Expert	3.13	1.18	893
	Culture_11	2.64	1.16	896
	Large Baseline	2.64	1.28	884
GPT	Culture_5	3.23	1.13	1063
	Culture_Expert	3.00	1.19	890
	Culture_11	2.88	1.16	882
	Large Baseline	2.64	1.22	889
Llama	Culture_5	2.50	1.03	900
	Culture_Expert	2.35	0.95	900
	Culture_11	1.97	0.93	900
	Large Baseline	2.29	1.06	900
Mistral	Culture_5	2.50	1.05	900
	Culture_Expert	2.30	0.99	900
	Culture_11	1.95	0.99	900
	Large Baseline	2.36	1.06	891
Qwen	Culture_5	2.63	0.99	900
	Culture_Expert	2.51	0.94	900
	Culture_11	2.26	0.99	899
	Large Baseline	2.83	1.23	900
Gemini	Culture_5	1.49	0.98	893
	Culture_Expert	1.45	0.86	888
	Culture_11	1.21	0.55	888
	Large Baseline	1.41	0.85	884

A.3 Complete Domain-Specific Effect Sizes for Novelty

Table 12 presents the complete domain-specific effect sizes for all persona-domain combinations.

Table 12: Complete Domain-Specific Effect Sizes for Novelty

Persona	Creative	General Knowledge	Instruction	Reasoning
Culture_5	0.207	0.495	0.159	0.453
Culture_Expert	0.032	0.174	0.079	0.175
Culture_11	-0.170	-0.118	-0.198	-0.124

Positive values indicate improvement over baseline; negative values indicate degradation.

A.4 Power Analysis Details

Table 13 presents the post-hoc power analysis results for each model-persona combination.

Table 13: Post-hoc Power Analysis for Novelty by Model and Persona

Model	Culture_5	Culture_Expert	Culture_11
Claude	0.95	0.82	0.78
DeepSeek	0.92	0.98	0.76
GPT	0.88	0.91	0.85
Llama	0.85	0.72	0.99
Mistral	0.98	0.81	1.00
Qwen	0.94	0.79	0.99
Gemini	0.79	0.68	0.94

Values > 0.80 indicate sufficient power. Mean power = 0.771; 71.4% of comparisons achieved sufficient power.

A.5 Leave-One-Out Sensitivity Results

Leave-one-out sensitivity analysis confirmed that results are highly stable. Table 14 presents the mean absolute change in effect size when each model was removed.

Table 14: Leave-One-Out Sensitivity Analysis for Novelty

Removed Model	Mean $ \Delta $ in Effect Size
Qwen	0.069
Claude	0.054
GPT	0.042
DeepSeek	0.041
Mistral	0.031
Gemini	0.021
Llama	0.016

Mean absolute change in Cohen’s d when each model is removed. All changes $|\Delta| < 0.1$, indicating high stability. Qwen and Claude have the largest impact, while Llama has the smallest. No single model drives the observed patterns.

A.6 Model Costs Summary

Table 15 summarizes the output token costs per 1M tokens for all models used in the economic analysis.

Table 15: Model Costs Summary (Output Token Cost per 1M Tokens)

Model	Small Model	Large Model	Cost Ratio (Small/Large)
Claude	\$15.00 (Sonnet)	\$25.00 (Opus)	0.60
DeepSeek	\$0.42 (Chat)	\$0.42 (Reasoner)	1.00
GPT	\$2.00 (Mini)	\$15.00 (5.4)	0.13
Llama	\$0.20 (8B)	\$0.88 (70B)	0.23
Mistral	\$0.60 (Mixtral)	\$0.30 (Small)	2.00
Qwen	\$0.30 (7B)	\$1.50 (80B)	0.20
Gemini	\$1.50 (Flash Lite)	\$12.00 (Pro)	0.13

Note: For Mistral, the "small" model (Mixtral) is actually older and more expensive than the "large" model (Mistral-Small), reflecting our architectural comparison design.

A.7 Results for Other Dimensions (Exploratory)

Table 16 presents the mean scores for Usefulness, Flexibility, Elaboration, and Cultural Sensitivity. These results are reported as exploratory due to poor inter-rater reliability (see Section 3.6).

Table 16: Mean Scores for Other Dimensions (Exploratory)

Model	Condition	Usefulness	Flexibility	Elaboration	Cultural Sensitivity
Claude	Culture_5	4.38	4.31	4.15	4.12
	Culture_Expert	4.35	4.18	4.08	4.05
	Culture_11	4.28	4.05	3.98	3.92
	Baseline	4.20	4.02	3.95	3.88
DeepSeek	Culture_5	4.28	4.22	4.05	3.98
	Culture_Expert	4.32	4.28	4.12	4.05
	Culture_11	4.18	3.95	3.85	3.78
	Baseline	4.05	3.88	3.75	3.68
GPT	Culture_5	4.22	4.15	4.02	3.95
	Culture_Expert	4.25	4.18	4.05	3.98
	Culture_11	4.18	4.08	3.95	3.88
	Baseline	4.12	3.98	3.85	3.78
Gemini	Culture_5	1.60	1.49	3.42	3.80
	Culture_Expert	1.55	1.45	3.35	3.75
	Culture_11	1.48	1.38	3.28	3.65
	Baseline	1.52	1.42	3.32	3.70

These results should be interpreted with caution due to poor inter-rater reliability (Fleiss' κ : Usefulness = 0.15, Flexibility = 0.30, Elaboration = 0.30, Cultural Sensitivity = -0.23).

A.8 Pure Persona Effects (Small Baseline vs. Small+Persona)

This supplementary analysis isolates the pure effect of each persona by comparing small models with personas against their own small model baseline (no persona). While our main analysis (Section 4) addresses the economic question of whether small+persona can match large model performance at lower cost, this analysis answers a different question: *does adding a persona improve the small model over its own baseline?*

Positive Cohen's d values indicate that the persona improved Novelty relative to the small baseline. Negative values indicate the persona harmed performance. All comparisons use Novelty scores (the only dimension with good inter-rater reliability, $\kappa = 0.68$). Statistical significance is determined using FDR-corrected $p < 0.05$ (indicated by *).

Table 17: Pure Persona Effects (Small Baseline vs. Small+Persona) for Novelty

Model	Persona	Baseline Mean	Persona Mean	Cohen's d
Claude	Culture 5	2.81	3.81	0.922
Claude	Culture Expert	2.81	3.13	0.263
Claude	Culture 11	2.81	2.88	0.062
DeepSeek	Culture 5	2.90	3.34	0.379
DeepSeek	Culture Expert	2.90	3.13	0.194
DeepSeek	Culture 11	2.90	2.64	-0.221
GPT	Culture 5	2.76	3.23	0.395
GPT	Culture Expert	2.76	3.00	0.199
GPT	Culture 11	2.76	2.88	0.099
Llama	Culture 5	2.22	2.50	0.279*
Llama	Culture Expert	2.22	2.34	0.125*
Llama	Culture 11	2.22	1.97	-0.272*
Qwen	Culture 5	2.35	2.63	0.281*
Qwen	Culture Expert	2.35	2.51	0.165*
Qwen	Culture 11	2.35	2.26	-0.091
Mistral	Culture 5	2.24	2.50	0.255*
Mistral	Culture Expert	2.24	2.29	0.057
Mistral	Culture 11	2.24	1.95	-0.286*
Gemini	Culture 5	1.41	1.49	0.092
Gemini	Culture Expert	1.41	1.45	0.051
Gemini	Culture 11	1.41	1.21	-0.277

*p < 0.05 (FDR-corrected). Positive Cohen's d indicates improvement from persona. Baseline = small model with no persona. Persona = small model with persona.

The pure persona effects are consistent with our main findings: Culture_5 consistently improves Novelty across most models (largest effect: Claude, $d = 0.922$), while Culture_11 shows negative or negligible effects for all models except Claude. The larger effect sizes observed here (compared to our main analysis) reflect the weaker baseline (small model without persona) rather than any inconsistency with our primary conclusions.

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